Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/05/2020
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Boiling: A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jyagoobi@wpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	508-831-5759
Organization Name:	Worcester Polytechnic Institute		
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PI Web Page:			
City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	3
Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/01/2016	**End Date:**	09/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	June 2019 report: No change to report.		
COI Name (Institution):	Didion, Jeffrey M.S. (NASA Goddard Space Flight Center)		
Grant/Contract No.:	NNX16AT09G		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation with same Principal Investigator (PI) Jamal Yagoobi, Ph.D. of "Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism," grant NNX12AR32G. Electrically Driven Liquid Film Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism is an International Space Station (ISS) Experiment to investigate the effects of gravity on: 1.) electrically generated two-phase flow and 2.) electrically driven liquid film boiling. The proposed work is a collaborative effort between the investigators to develop and validate the flight experiment hardware. We propose this effort based upon the successes of our current project: NASA Grant/Cooperative Agreement: NNX12AR32G Electrically Driven Liquid Film Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism. Although the original effort extends through February 2017, we have completed tasks and achieved the technical and programmatic milestones of		

Task Description:	this project. That is, we have passed all the requirements of the Science Concept Review and received the authorization to advance to the next stage of the experiment development. Therefore, the PI and Co-PI submit this proposal to develop the flight experiment hardware with an effective start date of October 2016. This is a proposal to support the development of the experiment flight hardware for the International Space Station, conduct the experiment on–board the International Space Station, support the operations of the experiment and evaluate the experimental data. The proposed period of performance is October 2016 to September 2021. The robust collaboration between Worcester Polytechnic Institute (WPI) and Goddard Space Flight Center (GSFC) can provide both state of the art theoretical investigation and deliver flight qualified experimental hardware; thus, this proposal presents both investigative and hardware development tasks. The PI will maintain overall technical direction of the experiment development; however, the Co-I will support the flight experimental hardware development in collaboration with Glenn Research Center. Note: This project successfully passed all requirements of the Science Concept Review and has received authorization to advance to the next stage of development. This is to elevate it to the International Space Station flight status as of October 1, 2016, with this continuation grant.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are a number of potential terrestrial and space applications for this electrohydrodynamic (EHD) driven two-phase heat transport device in all scales (from macro-scale to truly micro-scales). For example, this device can be used to effectively remove heat from electronics in micro- and zero-gravity environment for NASA's future space stations and long term missions. It can also be used in satellites (e.g., embedded system into satellite's multi-functional structure). Terrestrially, it can be used in high intensity electronics cooling as well as in various phase change devices.
Task Progress:	The following tasks have been accomplished during the period of October 2019 to July 2020. 1- Preliminary Design Review and Science Review Documents: The Preliminary Design Review was successfully carried out on May 27, 2020. The Science Review Document was further revised and baselined for the Experiment. This included working fluid change from HFE-7000 to HFE-7100. 2- Investigation of Gravity Effects on Electrically Driven Liquid Film Boiling: A Micro-Gravity Flight Campaign in Preparation of ISS Flight Experiment Variable gravity flight testing was conducted for a total of 120 parabolas over the course of three days during the week of November 17th, 2019, at Sanford International Airport outside of Orlando, Florida. Parabolic maneuvers were performed in sets of five, with each parabola giving approximately 15-20 seconds of microgravity. Even in the highly transient environment of a variable gravity flight, the results show that the electrohydrodynamic (EHD) conduction pumping and dielectrophoretic (DEP) electrode can provide similar enhancements to liquid film flow boiling in microgravity to those obtained in Earth gravity. 3- Thermal Analysis: Electrically Driven Thin Film Liquid Boiling in the absence of Gravity – Experiment Chamber: The thermal analysis considers the Engineering Test Unit (ETU) configuration for the Electrically Driven Thin Film Liquid Boiling in the absence of Gravity fabricated at the Goddard Space Flight Center. The analyses employ COMSOL Finite Element Analysis (FEA) software version 5.4. There are two analyses categories: 1.) nominal experiment operations 2.) loss of cooling and heater control.
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Abstracts for Journals and Proceedings	Castaneda AJ, O'Connor NJ, Yagoobi J. "Experimental Study of Flexible Electrohydrodynamic Conduction Pumping for Electronics Cooling." InterPACK2019. ASME 2019 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems, Anaheim, CA, October 7-9, 2019. InterPACK2019. ASME 2019 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems, Anaheim, CA, October 7-9, 2019. InterPACK2019-6746. , Oct-2019
Articles in Peer-reviewed Journals	O'Connor NJ, Castaneda AJ, Christidis PN, Tobar NV, Talmor M, Yagoobi JS. "Experimental study of flexible electrohydrodynamic conduction pumping for electronics cooling." Journal of Electronic Packaging. 2020 Dec;142(4):041105. Published Online: June 23, 2020. https://doi.org/10.1115/1.4047459 , Dec-2020
Articles in Peer-reviewed Journals	Vazquez PA, Talmor M, Seyed-Yagoobi J, Traore P, Yazdani M. "In-depth description of electrohydrodynamic conduction pumping of dielectric liquids: Physical model and regime analysis." Physics of Fluids. 2019 Nov;31(11):113601. https://doi.org/10.1063/1.5121164 , Nov-2019